

Territorial Information Retrieval from Heterogeneous Open Data through the Construction of a Data Warehouse for Water Management in Mexico City

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Abstract. This work presents a data warehouse for the territorial information retrieval of water consumption in Mexico City (CDMX), built from heterogeneous open sources. The contribution lies in [three](#) methodological elements: an ETL process that consolidates official SACMEX data into a unified dimensional model, [a declarative mapping that publishes the resulting star schema as an RDF knowledge graph aligned with the RDF Data Cube and GeoSPARQL vocabularies](#), so that territorial retrieval can also be expressed as graph traversal, and a deployment methodology that guarantees an accessible demonstration through a static version published on GitHub Pages and backed by an updated public repository. The prototype organises the information in a dimensional model and exposes it through a retrieval interface with an interactive choropleth map; filters by year, bimester, borough (*alcaldía*) and neighbourhood (*colonia*); and an exploratory module for detecting atypical consumption. The analysis reveals territorial and temporal consumption patterns, turning dispersed open data into queryable information for urban diagnosis, academic research, and citizen monitoring.

Keywords: Data warehouse · Information retrieval · Open data · ETL · Knowledge graph · RDF Data Cube · GeoSPARQL · Water consumption · Mexico City · Geospatial analysis

1 Introduction

Mexico City faces a water crisis characterised by aquifer overexploitation, leaks in the distribution network, and population growth that increases demand. Although open data on water consumption exist (SACMEX), they are published in poorly accessible formats and without integrated query tools, which limits their use for evidence-based decision making.

This work addresses the problem from two perspectives proper to information processing: the extraction of structured information from heterogeneous open sources through an ETL process, and the retrieval of that information through a parameterised query interface. The purpose is to transform dispersed datasets into a queryable dimensional repository that allows retrieving, filtering, and geospatially visualising water consumption by neighbourhood and borough, so that anyone can explore it.

The contribution of this work is [threefold](#): (i) a reproducible ETL process to consolidate official open data into a dimensional data warehouse; (ii) a [declarative mapping that lifts the consolidated star schema to an RDF knowledge graph](#), reusing the [RDF Data Cube](#) vocabulary for the observations and [GeoSPARQL](#) for the territorial geometries, which makes the same territorial retrieval [expressible both as SQL aggregation and as graph traversal](#); and (iii) a continuous-deployment methodology that keeps a functional demonstration always accessible, combining a dynamic deployment connected to a database with a static version published in a public repository.

The analysis of the 70,886 consolidated records reveals that the boroughs of Cuauhtémoc and Miguel Hidalgo lead the total consumption (Table 6), and that consumption varies across the available bimesters without a direct causal relationship with dry or rainy periods. [Because the source separates domestic, non-domestic and mixed billing, this ranking must be read as an aggregate of residential and commercial demand rather than as a per-capita indicator](#); [Section 5.2 develops this reading](#). Potential uses of the system include territorial diagnosis by citizens wishing to know the consumption of their own neighbourhood, prioritisation of inspection by the utility, statistical analysis by researchers, data journalism, and monitoring of water policy by civil organisations.

The remainder of the article is organised as follows: Section 2 reviews related work; Section 3 describes the methodology; [Section 4 presents the mapping from the dimensional model to a knowledge graph](#); Section 5 presents the results and evaluation; [Section 6 discusses the findings against previous systems and states the threats to validity](#); and Section 7 sets out the conclusions and future work.

2 Related Work

2.1 Foundations: Data Warehouses, Dimensional Modelling, and ETL

Dimensional modelling and the star schema constitute the consolidated basis for the design of analytical data warehouses. Kimball and Ross [1] establish the canonical principles —declaration of the grain, fact and dimension tables, surrogate keys, slowly changing dimensions, and the ETL subsystems— that ground the model adopted in this work. On the integration phase, Vassiliadis [2] offers a reference survey of *Extract–Transform–Load* technology, covering the conceptual and logical modelling of the processes and the cross-cutting extraction, transformation, and loading issues that methodologically support our pipeline.

2.2 Data Warehouses and Information Systems for Water Management

In the water domain, the most direct antecedent is that of Soares et al. [3], who implement a data warehouse with an ETL process to monitor water supply and consumption, integrating meter readings, reservoir volumes, and geographic location from management-application databases and heterogeneous files. Duque-Méndez, Orozco-Alzate, and Vélez [4] develop a star schema with multidimensional analysis (OLAP) for historical hydro-climatological series, with emphasis on data-quality assessment. Abecker et al. [5] present a hybrid sensor-semantic architecture (the WatERP project) that combines a relational/PostGIS store with a triple store for integrated resource management. These systems, however, mostly start from operational data, SCADA, or proprietary sensor networks, and not from open government data.

2.3 Open Government Data: Integration, Quality, and Reuse

Integrating open government data into dimensional warehouses poses its own challenges. Berro, Megdiche, and Teste [7] propose “content-driven” ETL processes that address precisely the semantic and structural heterogeneity, the absence of schemas, and the dispersion characteristic of open data, factors that overwhelm traditional ETL. Complementarily, Vetrò et al. [6] define a quantitative measurement framework for the quality of Open Government Data (accuracy, completeness, consistency, traceability, and freshness) and empirically report that the most frequent problems are missing metadata, incomplete data, and the absence of update information —exactly the challenges our ETL faces over the SACMEX data.

More recent evidence confirms that the bottleneck has shifted from publication to reuse. Quarati [14] analyses usage trends across open data portals and shows that a large share of published datasets remains effectively unused, with metadata quality acting as a strong predictor of uptake. Nikiforova and McBride [15] reach a convergent conclusion from a user-centred usability analysis of 41 government portals: the obstacle is rarely availability, but the absence of tooling that turns a downloadable file into an answerable question. Weil et al. [16] survey urban digital-twin initiatives and identify fragmented, non-interoperable municipal data as the dominant barrier to sustainable smart-city analytics. These three results motivate the design decision at the core of this work: rather than publishing yet another dataset, we publish a queryable and permanently reachable retrieval layer over data that already exists.

2.4 Geospatial Information Retrieval and Visualisation

Purves et al. [8] offer a reference monograph on Geographic Information Retrieval (GIR) —indexing, search, and spatial ranking— that theoretically frames the territorial information-retrieval component of this work. On visualisation, Han et al. [9] present an open-source web tool for exploring choropleth maps with

automated class intervals and synchronous linking, a technique that directly supports our interactive choropleth visualisation by borough.

2.5 Knowledge Graphs for Urban and Water Domains

A parallel line of work represents territorial information as a graph rather than as a cube. GeoSPARQL, in its 1.1 revision [19,17], standardises classes, properties, and topological functions for querying geospatial RDF, and Jovanovik et al. [18] provide a compliance benchmark showing that mainstream triple stores now implement the core of the standard. On that basis, WorldKG [23] lifts OpenStreetMap into a world-scale geographic knowledge graph, UrbanKG [24] organises heterogeneous city data into a reusable urban knowledge graph, and UrbanFloodKG [25] applies the same idea to urban flood-risk assessment, a domain adjacent to ours.

In the water domain specifically, Roussey et al. [26] argue that knowledge-graph technologies are the natural next step for food, agriculture and water data, precisely because these domains combine many small, heterogeneous, authority-owned sources. Papageorgiou et al. [27] build a knowledge graph over industrial water-treatment processes, and Ramonell et al. [28] use graph-based integration for digital twins of built assets. What these systems share with ours is the integration problem; what they do not share is the starting point, since they operate over sensor, process or cadastral data rather than over municipal open data published as flat files.

For the construction step, R2RML [21] is the W3C recommendation for declaratively mapping relational databases to RDF, and RML [22] generalises it to heterogeneous formats. The RDF Data Cube vocabulary [20] provides the standard representation for multidimensional statistical observations, which makes it the natural target for a star schema. Section 4 exploits exactly this correspondence.

2.6 Reproducibility and Deployment of Scientific Demonstrations

Sandve et al. [10] formulate widely cited rules for computational reproducibility (version control, documentation of preprocessing, public access to data and scripts) that ground our reproducible static version. Hill et al. [11] provide a practical guide to developing and deploying an open-source, reproducible web dashboard for water-sanitation data, a close domain that supports the continuous-deployment methodology with reproducible artefacts.

2.7 Positioning of the Contribution

Table 1 contrasts the present work with the closest systems along the four axes that matter for this problem: the nature of the input, the data model, the retrieval interface exposed to the end user, and whether a permanently reachable artefact is provided.

Table 1. Comparison with the closest related systems. “OGD” denotes open government data; “perm. demo” denotes a demonstration that remains reachable after the evaluation period.

System	Input	Model	Retrieval	Perm. demo
Soares et al. [3]	Utility DBs, meters	Star schema	OLAP	No
Duque-M. et al. [4]	Hydro-clim. series	Star schema	OLAP	No
Abecker et al. [5]	SCADA, sensors	Rel.+RDF	SPARQL	No
Berro et al. [7]	OGD (generic)	Star schema	OLAP	No
UrbanKG [24]	Mixed city data	RDF graph	Graph queries	Dataset
UrbanFloodKG [25]	Flood data	RDF graph	Graph queries	No
Hill et al. [11]	Wastewater data	Tabular	Dashboard	Yes
This work	SACMEX OGD (CSV)	Star + RDF	REST + SPARQL	Yes

Unlike existing water data warehouses—which start from operational data, SCADA, or proprietary sensors [3,4,5]—and recognising that the integration of heterogeneous open data remains an unsolved problem in this domain [7,6,14], this work differs in four respects: an ETL process that consolidates official SACMEX open data into a dimensional star schema; the exposure of that information through a REST API and a choropleth visualisation supporting territorial retrieval [8,9]; a declarative R2RML mapping that republishes the same facts as an RDF knowledge graph aligned with RDF Data Cube and GeoSPARQL, placing the resource in the Linked Open Data space and enabling topological queries the relational interface does not naturally express; and a continuous-deployment methodology whose reproducible static version guarantees indefinite verifiability [10,11]. The same approach was previously applied to the seismic-risk domain in Mexico [12], a direct methodological antecedent transferred here to the water domain. To the best of our review—over Scopus, Web of Science, IEEE Xplore, the ACM Digital Library and Google Scholar, combining the terms *data warehouse*, *knowledge graph*, *open data*, *water consumption*, *SACMEX* and *Mexico City*—no peer-reviewed, reproducible data warehouse built specifically over SACMEX/CDMX open data exists.

3 Methodology

3.1 General Architecture

The system follows a layered architecture: a data layer (PostgreSQL), a processing layer (ETL with Python and SQL), an API layer (FastAPI), and a presentation layer (HTML/CSS/JS frontend with Leaflet and Plotly). The warehouse implements a star schema with one fact table (`fact_water_consumption`) and three dimensions: `dim_time`, `dim_location` (borough, neighbourhood, latitude, longitude), and `dim_dev_index` (development level). Following Kimball

and Ross [1], `dim_time` is a full-date dimension: each row is a distinct reading date present in the source, with the derived attributes `date`, `year`, `month`, `day` and `bimester`. The bimester is therefore a derived attribute of the date and not the key of the dimension, which explains the cardinality reported in Section 5.1.

3.2 Data Source and Grain of the Fact Table

The source is the *Consumo de agua* dataset published by SACMEX through the open-data portal of Mexico City [13]. It reports bimonthly billed water supply, disaggregated by territorial unit, and distinguishes three billing categories—domestic, non-domestic and mixed—each with a total and an average consumption expressed in cubic metres. It also carries a development index (*índice de desarrollo*) with four ordinal levels, which is published as part of the dataset rather than derived by us.

Declaring the grain is the decisive modelling step [1], and we state it explicitly: one row of `fact_water_consumption` corresponds to one **[confirm the exact unit: manzana (city block) or colonia]** of the source, for one billing period. The measures stored at that grain are the total and average consumption for each of the three billing categories, in cubic metres. Because several territorial units belong to the same neighbourhood, `dim_location` is coarser than the grain, and every aggregate reported in Section 5 is therefore a roll-up over the fact rows rather than a direct reading of a source field.

3.3 ETL Consolidation Process

The first contribution of the work is the ETL process that consolidates the SACMEX open data into the dimensional warehouse. The complete detail of the process—the SQL schema (DDL), the `loader.py` loading script, and the parameterised retrieval queries—is documented in the project’s public repository; this article presents only the architecture of the process (Fig. 1).

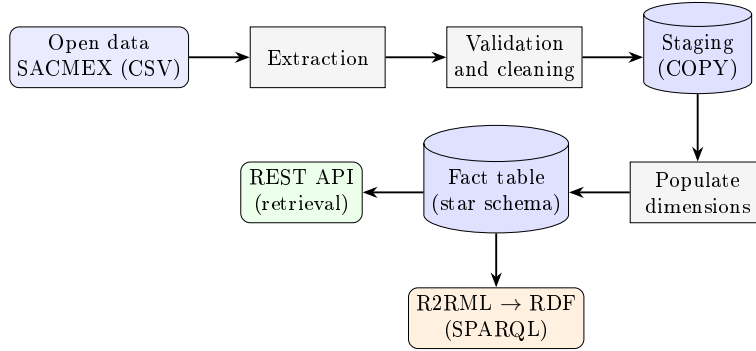
The pipeline operates in four phases. (1) **Extraction**: reading the open SACMEX CSV files. (2) **Validation and cleaning**: discarding records with empty fields, inconsistent coordinates, or incompatible formats. (3) **Loading to staging and populating dimensions**: COPY of the data into temporary tables and extraction of the unique values into `dim_time`, `dim_location`, and `dim_dev_index`. (4) **Populating the fact table**: a join between staging and dimensions to populate `fact_water_consumption` under the star schema, finally dropping the staging tables. The information thus consolidated becomes available for retrieval through parameterised queries exposed by the REST API.¹

The exclusion criteria applied in phase (2) are deterministic and were fixed before loading, so that the pipeline is idempotent: a record is discarded if any

¹ SACMEX operates a proprietary industrial-grade data platform (commercial technology introduced in 2021) that integrates SCADA and multiple sources; it is distinguished from a reproducible academic work over open data such as the present one.

Table 2. Records discarded during the validation and cleaning phase.

Exclusion criterion	Records % of source	
Missing or non-numeric measure	[?]	[?]
Null or out-of-range coordinates	[?]	[?]
Unmatched territorial label	[?]	[?]
Unparseable reading date	[?]	[?]
Total discarded	[?]	[?]
Loaded into the fact table	70,886	[?]

**Fig. 1.** ETL process architecture: from SACMEX heterogeneous open data to the star schema exposed through a REST API and, through a declarative R2RML mapping, as an RDF knowledge graph.

of the consumption measures is absent or non-numeric, if the geographic coordinates fall outside the bounding box of Mexico City or are null, if the borough or neighbourhood label cannot be normalised against the official catalogue, or if the reading date cannot be parsed. Table 2 reports the volume affected by each criterion, following the recommendation of Vetrò et al. [6] to quantify rather than merely to declare data quality.

3.4 Frontend and Geospatial Visualisation

The frontend offers a dashboard with indicators, interactive charts (Plotly), and a choropleth map of the 16 boroughs (Leaflet + GeoJSON), where colour represents the consumption level and each borough displays a popup with its detail (Fig. 2).

3.5 Implementation of the Atypical-Consumption Module

The current version replaces the idea of a traditional supervised prediction with an atypical-consumption detection module oriented to analytical retrieval. Instead of training a model with historical anomaly labels—which are not available in the open source—the system computes statistical indicators over the

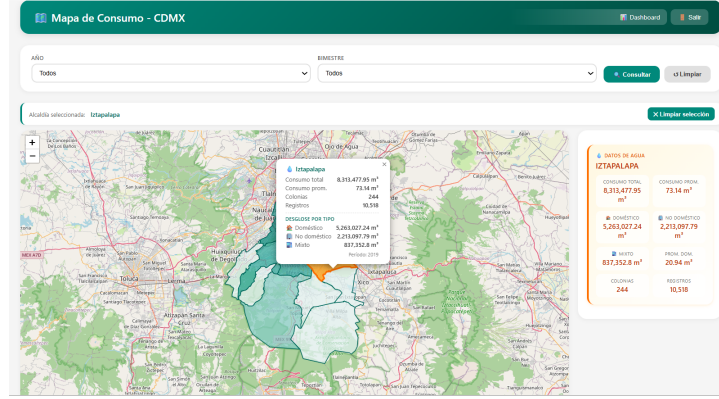


Fig. 2. Interactive map of water consumption by borough in Mexico City.

filtered records and detects zones whose consumption departs significantly from the aggregate behaviour.

The procedure is implemented as an unsupervised analysis on the frontend: first, the records are grouped by borough, neighbourhood, year, and bimester; then reference measures such as total consumption, average consumption, and relative dispersion are computed; finally, the zones with the largest deviation are ordered to show them as possible atypical consumption. This approach avoids presenting a prediction as if it were a model trained with reference data (ground truth), but keeps the practical value of the analysis component by flagging patterns that require later inspection.

Formally, for each territorial group g an atypicality score A_g is computed from the normalised deviation with respect to the mean of the filtered set:

$$A_g = \left| \frac{x_g - \mu}{\sigma + \epsilon} \right|, \quad (1)$$

where x_g represents the aggregate consumption of the group, μ is the average consumption of comparable groups —that is, of the groups that remain after applying the same filter combination selected by the user, which constitutes the reference population against which the deviation is measured—, σ is the standard deviation, and ϵ prevents division by zero. The groups with the largest A_g are reported in the interface as zones with atypical consumption.

Two clarifications are necessary to delimit what this module does and does not do. First, it is an *exploratory anomaly-detection* component and not a causal predictor of consumption: a high A_g states that a zone departs from its comparison set, not why. Second, and in direct response to a possible reading of the results in Section 5.4, the module does *not* perform real-time leak detection. No real leak reports, no work orders and no network telemetry are integrated into the warehouse, and the quantitative evaluation reported below was obtained exclusively over synthetically injected perturbations. Its operational role is to

prioritise zones for later inspection by the utility, which is a retrieval task — ranking territorial units by a computed score— and not a diagnostic one.

3.6 Continuous Deployment Methodology

To guarantee that the demonstration remains available and reproducible without depending on the availability of a server, the system is published in two complementary modalities. The **dynamic deployment** keeps the complete architecture —FastAPI over PostgreSQL— and serves parameterised queries and aggregations against the data warehouse; it was used to obtain the experimental results reported in this article, and its availability depends on the hosting service. The **static version** exports the consolidated data as JSON and GeoJSON files that the frontend consumes directly, so that querying, filtering, aggregation and visualisation work on the client without an active backend. The public repository is the preservation mechanism: it allows the static version to be reconstructed and executed independently and indefinitely. [This duality addresses the recurring practical problem that academic prototypes stop working shortly after evaluation, a concern also raised by Hill et al. \[11\].](#)

The system is available at the following addresses:

- **Public repository:** https://github.com/Uriel1024/Data_Warehouse_static
- **Static demo (permanent, no backend required):** https://uriel1024.github.io/Data_Warehouse_static/

The experimental results, the statistical analysis, and the findings reported in this article were obtained over the original data warehouse built from the official SACMEX open data, comprising 70,886 records. The static version is intended to show the operation of the information-retrieval interface and does not replace the database used for the experimental evaluation.

3.7 Technologies Used

The backend is implemented with FastAPI and Python 3.10 over Uvicorn; the warehouse runs on PostgreSQL 14 accessed through psycopg2; the frontend uses HTML5, CSS3, JavaScript, Leaflet and Plotly.js, consuming JSON and GeoJSON exports; the ETL is written in Python and SQL; and [the graph layer relies on an R2RML mapping targeting the RDF Data Cube and GeoSPARQL vocabularies, queried with SPARQL 1.1.](#)

4 From the Dimensional Model to a Knowledge Graph

4.1 Rationale: Why a Star Schema, and What a Graph Adds

The choice of a relational star schema as the primary store was not made by default, and it is worth justifying it explicitly. The dominant workload of the system

Table 3. Alignment between the dimensional model and the RDF vocabularies.

Relational element	RDF construct	Vocabulary
<code>fact_water_consumption</code> (row)	<code>qb:Observation</code>	RDF Data Cube
Measures (total, average, m ³)	<code>qb:MeasureProperty</code>	RDF Data Cube
<code>dim_time</code> , <code>dim_location</code>	<code>qb:DimensionProperty</code>	RDF Data Cube
Borough / neighbourhood entity	<code>geo:Feature</code> , <code>geo:asWKT</code>	GeoSPARQL
Territorial adjacency	<code>geo:sfTouches</code>	GeoSPARQL
<code>dim_dev_index</code>	<code>skos:Concept</code>	SKOS
Billing category (dom./non-dom./mixed)	<code>qb:AttributeProperty</code>	RDF Data Cube

is the repeated aggregation of a single numeric measure over three conformed dimensions —time, territory and development level— with filter combinations chosen interactively by the user. This is precisely the workload for which dimensional modelling was designed [1]: the star schema keeps the join depth constant at one, allows the aggregate to be computed by a single scan with grouping, and is directly supported by the indexing and query planning of a mature relational engine. A graph representation of the same data would pay a traversal cost for joins that the star schema resolves with foreign keys, and would not, by itself, accelerate the aggregation.

What a graph representation adds is complementary rather than competing: *interoperability*, since a standard vocabulary makes the resource linkable to other municipal datasets without prior schema agreement, the recurring obstacle identified in the open-data literature [14,15,16]; *topological expressiveness*, since questions such as “which neighbourhoods adjacent to a zone with atypical consumption also exceed the borough mean” are natural traversals over an adjacency relation and awkward over a flat territorial dimension; and *entity reconciliation*, since dereferenceable identifiers aligned with external authorities resolve the label heterogeneity that our ETL currently absorbs through normalisation rules. We therefore keep the star schema as the analytical store and derive an RDF layer from it by a declarative mapping, so that neither representation is maintained by hand.

4.2 Vocabulary Alignment and R2RML Mapping

The correspondence between a star schema and the RDF Data Cube vocabulary [20] is close to mechanical, which is what makes this layer inexpensive to maintain. Table 3 states the alignment adopted.

The mapping is expressed in R2RML [21], the W3C recommendation for transforming relational content into RDF; RML [22] would allow the same rules to be applied directly to the source CSV files, which is the variant we intend to adopt when the pipeline is generalised to further open-data domains. Listing 1.1 shows the triples map for the fact table; the complete mapping is distributed with the repository.

Listing 1.1. Excerpt of the R2RML triples map for the fact table.

```

@prefix rr: <http://www.w3.org/ns/r2rml#> .
@prefix qb: <http://purl.org/linked-data/cube#> .
@prefix geo: <http://www.opengis.net/ont/geosparql#> .
@prefix agua: <https://w3id.org/cdmx/agua/> .

<#ObservationMap> a rr:TriplesMap ;
  rr:logicalTable [ rr:tableName "fact_water_consumption" ] ;
  rr:subjectMap [ rr:template "https://w3id.org/cdmx/agua/obs/{fact_id}" ;
    rr:class qb:Observation ] ;
  rr:predicateObjectMap [ rr:predicate qb:dataSet ;
    rr:objectMap [ rr:constant agua:consumoCDMX ] ] ;
  rr:predicateObjectMap [ rr:predicate agua:location ;
    rr:objectMap [ rr:parentTriplesMap <#LocationMap> ] ] ;
  rr:predicateObjectMap [ rr:predicate agua:totalConsumption ;
    rr:objectMap [ rr:column "consumo_total" ;
      rr:datatype xsd:decimal ] ] .

<#LocationMap> a rr:TriplesMap ;
  rr:logicalTable [ rr:tableName "dim_location" ] ;
  rr:subjectMap [ rr:template "https://w3id.org/cdmx/agua/colonia/{location_id}" ;
    rr:class geo:Feature ] ;
  rr:predicateObjectMap [ rr:predicate geo:asWKT ;
    rr:objectMap [ rr:column "wkt" ;
      rr:datatype geo:wktLiteral ] ] .

```

4.3 Territorial Retrieval as Graph Traversal

The value of the layer is that it makes a class of territorial questions expressible that the tabular interface does not naturally support. GeoSPARQL 1.1 supplies the topological predicates [19,17], and current triple stores implement them at a level verified by the compliance benchmark of Jovanovik et al. [18]. Listing 1.2 gives the query for the adjacency example discussed above: it retrieves the neighbourhoods that border a zone flagged as atypical and whose own consumption exceeds the borough mean, a pattern that would require either a materialised adjacency table or a recursive query in the relational setting.

Listing 1.2. SPARQL query combining the statistical cube with the GeoSPARQL topology.

```

PREFIX qb: <http://purl.org/linked-data/cube#>
PREFIX geo: <http://www.opengis.net/ont/geosparql#>
PREFIX agua: <https://w3id.org/cdmx/agua/>

SELECT ?neighbour (SUM(?m) AS ?total) WHERE {
  agua:colonia/flagged geo:sfTouches ?neighbour .
  ?obs a qb:Observation ;
    agua:location ?neighbour ;
    agua:period ?p ;
    agua:totalConsumption ?m .
  ?p agua:bimester "2" .
}
GROUP BY ?neighbour
HAVING (SUM(?m) > 1739.40)
ORDER BY DESC(?total)

```

Table 4. Comparison of candidate data models for territorial retrieval of water consumption.

Requirement	Star schema	Property graph	RDF graph
Interactive aggregation over 10^5 rows	Very good	Moderate	Moderate
Multi-filter slicing (time, place, index)	Very good	Good	Good
Topological / adjacency queries	Weak (needs extra table)	Good	Very good (GeoSPARQL)
Linking to external datasets	Weak (schema agreement)	Moderate	Very good (IRIs)
Standard published vocabulary	No	No	Yes (Data Cube)
Schema evolution	Rigid	Flexible	Flexible
Operational cost for this prototype	Low	Medium	Medium

4.4 Comparison of Candidate Data Models

Table 4 summarises the trade-off that motivated the hybrid design, comparing the relational star schema, a labelled property graph, and an RDF knowledge graph against the requirements of this use case.

The conclusion, which we consider transferable to comparable municipal open-data projects, is that the two models answer different questions and that deriving the second from the first through a declarative mapping costs far less than maintaining either alone. Materialising the full graph into a triple store and comparing SPARQL against SQL over an equivalent workload are reported as ongoing work in Section 7.

5 Results and Evaluation

5.1 Integrated Data Volume

The ETL process consolidated the water-consumption information of Mexico City into the dimensional warehouse. Table 5 presents the stored volume. The consistency of the reported volumes was verified directly on the original data warehouse: the fact table contains 70,886 records, while the dimensions contain 1,553 locations, 181 time records, and 4 development-index levels. As stated in Section 3, the 181 rows of `dim_time` correspond to distinct reading dates present in the source and not to year–bimester combinations, and the 1,553 rows of `dim_location` are coarser than the fact grain.

5.2 Findings of the Consumption Analysis

The analysis of the 70,886 consolidated records reveals the following consumption patterns in Mexico City.

Table 5. Data volume stored in the data warehouse.

Table	Records
fact_water_consumption	70,886
dim_location	1,553
dim_time	181
dim_dev_index	4

Territorial concentration. The boroughs with the highest total consumption are Cuauhtémoc and Miguel Hidalgo, followed by Benito Juárez and Gustavo A. Madero (Table 6). This ordering requires a more careful reading than “the boroughs that consume the most”. Cuauhtémoc, Miguel Hidalgo and Benito Juárez concentrate the historic centre and the densest commercial activity of the city, so their totals aggregate a residential base with substantial non-domestic demand, whereas Gustavo A. Madero and Álvaro Obregón are among the most populous boroughs and reach comparable totals through volume of households. The warehouse supports the distinction directly, since the source separates domestic, non-domestic and mixed billing: the non-domestic share is [% Cuauhtémoc] against [% Gustavo A. Madero]. Total consumption is therefore not a per-capita indicator and should not be used as one for allocating supply interventions; the per-account average, also stored in the warehouse, is the appropriate measure.

Temporal distribution. Consumption shows variations across the bimesters available in the warehouse. In the aggregated query by bimester, the highest total consumption was observed in bimester 2, followed by bimester 3 and bimester 1 (Table 7). The spread between the extreme bimesters is below 10% of the mean, and the warehouse covers only three consecutive billing periods, which is insufficient to separate a seasonal signal from ordinary billing variation. Therefore, no direct causal relationship with dry or rainy periods is established without integrating additional climatic variables and at least two complete annual cycles.

Relationship with the development index. Neighbourhoods with a High development index show a higher average consumption than those with a Low index. The index is published as part of the source dataset rather than derived by us, and the portal does not document the methodology by which it was constructed; this is an instance of the missing-metadata problem quantified by Vetrò et al. [6] and by Quarati [14]. The observation is therefore descriptive and internally consistent, but attributing it to a socioeconomic mechanism would require an externally documented index, and we report it as an association rather than as a causal relationship.

The interface dynamically generates the list of neighbourhoods with the highest total consumption according to the filters selected by the user, so that the analytical outputs of the system —top neighbourhoods, total consumption by borough, average consumption per record, and the atypical-consumption ranking— are always relative to the active query rather than fixed in advance.

Table 6. Five boroughs with the highest total water consumption (m^3 , aggregated over the three available bimesters).

Borough	Total (m^3)
Cuauhtémoc	18,401,765.79
Miguel Hidalgo	14,490,397.15
Benito Juárez	13,780,118.64
Gustavo A. Madero	13,480,100.57
Álvaro Obregón	9,608,545.20

Table 7. Total and average consumption by bimester (m^3 ; average per fact record).

Bim.	Avg.	Total
1	1,618.82	37,663,466.69
2	1,739.40	41,519,580.96
3	1,733.65	41,174,097.68

5.3 The Atypical-Consumption Module within the Retrieval Framework

The atypical-consumption module makes it possible to order boroughs or neighbourhoods according to their deviation from the average behaviour of the filtered set. Its role within the overall framework is best understood as a ranking function rather than as an analytical add-on. The choropleth map answers “how much is consumed here”; the module answers “which territorial units should I look at first”, which is the ordering problem at the centre of information retrieval [8]. It consumes the same filtered result set that the retrieval interface produces, and returns a permutation of it, so that the user moves from browsing to a prioritised list without changing the query. This is what turns a descriptive visualisation into an exploration tool: the score is not a diagnosis, it is a relevance criterion for territorial units.

5.4 Quantitative Evaluation of the Atypical-Consumption Module

Although the module operates in an unsupervised manner in production —the open source provides no anomaly labels— its performance was evaluated through a controlled and reproducible protocol with injected anomalies acting as ground truth. Over a synthetic set with the structure of the warehouse ($N \approx 11,460$ records from 16 boroughs), 3 % of labelled anomalies were injected (70 % high spikes, 30 % drops) and three detectors were compared: the A_g z-score of this work, Isolation Forest, and LOF, using as features the log-scaled consumption and its ratio to the neighbourhood mean. Table 8 reports precision, recall, F1, and the areas under the ROC and precision–recall curves.

The z-score reaches perfect precision but moderate recall (0.50): it reliably detects extreme spikes, although its fixed threshold misses part of the drops. The asymmetry follows from the shape of the data rather than from the statistic itself: consumption is bounded below by zero, so an injected drop produces a smaller absolute deviation than an injected spike of the same relative magnitude, and a symmetric threshold at $A_g > 3$ therefore captures the spikes while leaving part of the drops below the cut. Isolation Forest offers the best trade-off ($F1 = 0.97$), while LOF proves poorly suited to this type of global-magnitude anomaly ($F1 =$

Table 8. Evaluation of the atypical-consumption detection module on injected anomalies (synthetic ground truth; $N \approx 11,460$, 3 % anomalies, fixed seed). Reproducible with `evaluacion-ml/eval_anomalias.py`.

Method	Precision	Recall	F1	ROC-AUC	PR-AUC
z-score A_g (this work)	1.000	0.497	0.664	0.987	0.816
Isolation Forest	0.940	1.000	0.969	1.000	1.000
LOF	0.135	0.029	0.048	0.555	0.062

0.05). These results support the use of the z-score as a high-precision exploratory filter and point to Isolation Forest as the natural improvement of the module.

Validity of the protocol. These metrics do not represent performance against real leaks or observed anomalies, but rather the ability of each method to recover injected synthetic anomalies under a controlled protocol. This decision is due to the fact that the open source does not provide real anomaly labels. Three limitations of the protocol must be stated. First, the features used —log-scaled consumption and ratio to the neighbourhood mean— are aligned with the mechanism by which the anomalies were injected, which inflates the separability of the problem and explains the perfect areas under the curve obtained by Isolation Forest; the figures should be read as an upper bound, not as expected field performance. Second, the thresholds were not equalised across methods: the z-score used the fixed statistical criterion $A_g > 3$, while Isolation Forest used $s > 0.60$ and LOF $s > 1.5$, so the comparison at a single operating point favours the methods whose threshold was chosen. Third, and as stated in Section 3.5, none of this constitutes evidence about real-time leak detection, for which the system is not designed and for which no labelled data were available. Therefore, the results should be interpreted as a reproducible evaluation of sensitivity to known perturbations, not as a definitive operational validation.

5.5 Performance Metrics

The performance of the main dashboard was evaluated using Lighthouse (v13.2.0). The measurement was taken against the dynamic deployment serving the complete warehouse, with the dashboard performing its initial render and its first aggregate request; the client-side aggregation of the exported dataset in the static version was measured separately and is not mixed into these figures. Table 9 summarises the metrics obtained.

Query execution over the total data volume. Frontend metrics do not characterise the retrieval layer, so the response time of the analytical endpoints was measured directly against the warehouse. Three workloads were distinguished: a paginated retrieval that returns a bounded page of facts, a full-scan aggregation by borough that touches all 70,886 rows, and a full-scan aggregation by neighbourhood and bimester, which is the heaviest query the dashboard issues. Each workload was executed [n] times after a warm-up round, on [hardware/hosting description], with the results reported in Table 10. Reporting

Table 9. Dashboard performance metrics (Lighthouse v13.2.0).

Metric	Value Assessment	
First Contentful Paint (FCP)	0.8 s	Excellent
Largest Contentful Paint (LCP)	1.2 s	Good
Speed Index	1.5 s	Excellent
Total Blocking Time (TBT)	0 ms	Excellent
Cumulative Layout Shift (CLS)	0	Excellent

Table 10. Response time of the retrieval endpoints over the complete warehouse (70,886 fact rows).

Workload	Rows scanned	Mean (ms)	p95 (ms)	Executions
Paginated retrieval (limit=100)	100	24.9	43.6	10
Full aggregation by borough	70,886	[?]	[?]	[?]
Aggregation by neighbourhood and bimester	70,886	[?]	[?]	[?]
Top-10 neighbourhoods with ordering	70,886	[?]	[?]	[?]

the 95th percentile over a sample of this size, rather than over the ten executions of a single endpoint used in the submitted version, is what makes the percentile meaningful.

5.6 Functional Tests and Retrieval Capability

The operation of the ETL processes, the persistence in PostgreSQL, the FastAPI endpoints, and the information retrieval was verified. The tests included queries parameterised by year, bimester, borough, and neighbourhood; aggregated retrieval (total and average consumption by neighbourhood); generation of JSON responses; geospatial retrieval for visualisation; and pagination with configurable limits. All modules produced results consistent with the expected structure. The dimensional model thus allows retrieving territorial information through combined filters of time, location, and development index, executing analytical queries over 70,886 records without directly accessing the original sources.

6 Discussion

6.1 Comparison with Previous Systems

Placed against the systems summarised in Table 1, the contribution of this work is neither the dimensional model nor the dashboard, both of which are well-established techniques, but the combination of an input that the water-management literature has largely avoided with an output that the open-data

literature has repeatedly asked for. The water data warehouses of Soares et al. [3] and Duque-Méndez et al. [4] obtain higher-quality input than ours—meter readings and instrumented series—at the cost of depending on the utility’s internal systems, which makes them non-replicable by third parties and non-auditable by citizens. Abecker et al. [5] anticipate our hybrid relational/RDF arrangement, but again over proprietary sensor networks. Conversely, the urban knowledge graphs of Liu et al. [24] and Wang et al. [25] operate on open input and publish reusable artefacts, but target prediction and risk assessment rather than the retrieval of billed consumption at territorial granularity.

The position we occupy is therefore the intersection: official open data as the only input, a dimensional model for the aggregation workload, a standard graph vocabulary for interoperability, and a permanently reachable demonstration. The closest precedent on the last axis is Hill et al. [11], whose wastewater-surveillance dashboard shares our concern with survivability, and whose experience motivated our decision to export a self-contained static version rather than to rely on a hosted backend.

6.2 Practical Implications

Three implications follow from the results. For the utility, the ranking produced by the atypical-consumption module is a prioritisation aid whose value lies in reducing the search space for inspection, not in diagnosing causes; used together with the domestic / non-domestic decomposition of Section 5.2, it separates zones whose deviation is plausibly commercial from those where a residential anomaly warrants a visit. For urban research, the consolidated warehouse removes the preprocessing barrier that currently makes this dataset expensive to use, and the graph layer of Section 4 lowers the cost of joining it with other municipal sources. For open-data policy, the exercise is a concrete measurement of the gap identified by Quarati [14] and Nikiforova and McBride [15]: the data were public throughout, yet a non-trivial engineering effort was required before a citizen could answer a question as simple as how much water is billed in their neighbourhood.

6.3 Threats to Validity and Limitations

Several limitations delimit the scope of the conclusions. The most consequential concerns temporal coverage: the warehouse holds three consecutive billing periods, which supports territorial comparison but not seasonal or trend analysis, and any reading of the bimester differences as a seasonal effect would be unwarranted. A second limitation concerns the development index, which is consumed as published and whose construction methodology is not documented by the source, so the association reported in Section 5.2 cannot be promoted to a socioeconomic explanation. A third concerns the evaluation of the detection module, which rests on synthetic perturbations whose feature signature is aligned with the detectors, and which therefore establishes sensitivity to known deviations rather than field performance against real leaks. A fourth concerns

the published static version, which serves exported JSON and GeoJSON files and consequently does not execute SQL against the warehouse in real time; the analytical results reported here were obtained on the dynamic deployment. A fifth concerns the graph layer, which is delivered as a declarative mapping and a set of example queries rather than as a materialised and benchmarked triple store. Finally, the quality of the whole rests on the quality, coverage and freshness of the source dataset, whose update cadence is outside our control and whose metadata gaps we absorb but cannot repair.

7 Conclusions and Future Work

7.1 Conclusions

A data warehouse was presented for the territorial information retrieval of water consumption in Mexico City, built from open sources through a reproducible ETL process and published with a continuous-deployment methodology that guarantees a permanent functional demonstration. The solution consolidated 70,886 consumption records and 1,553 locations across the 16 boroughs into a dimensional star schema on PostgreSQL. On top of that schema, a declarative R2RML mapping aligned with the RDF Data Cube and GeoSPARQL vocabularies republishes the same facts as a knowledge graph, so that territorial retrieval can be expressed both as relational aggregation and as topological traversal without duplicating the maintenance effort.

The findings reveal that water consumption in Mexico City is concentrated in the boroughs of Cuauhtémoc and Miguel Hidalgo (Table 6), a concentration that the domestic / non-domestic decomposition attributes to commercial intensity rather than to residential demand alone, and that it varies across the available bimesters by a margin too small, and over a window too short, to support a seasonal interpretation. These results show that the system not only consolidates data, but also allows extracting knowledge useful for water management, while making explicit the boundary beyond which the available data do not license further inference.

The three central contributions —the consolidation ETL, the graph layer derived from the dimensional model, and the always-accessible demo methodology— constitute a replicable method for bringing heterogeneous official open data to queryable, linkable and verifiable territorial information.

7.2 Future Work

The most immediate line of work is the materialisation of the knowledge graph described in Section 4 into a triple store, together with a quantitative comparison of SPARQL and SQL response times over an equivalent workload and the reconciliation of borough and neighbourhood entities against external authorities, which would place the resource in the Linked Open Data cloud and allow it to be joined with other municipal datasets without prior schema agreement. In

parallel, the temporal coverage of the warehouse must be extended to at least two complete annual cycles before any seasonal analysis is attempted, and climatic variables integrated if the relationship with dry and rainy periods is to be tested rather than merely observed.

The evaluation of the atypical-consumption module will be recomputed when real labelled anomaly or leak data become available, reporting complete precision–recall curves and a threshold sensitivity analysis at equalised operating points, and replacing the current synthetic protocol whose optimism we have made explicit. Finally, the method is domain-agnostic by construction, and we intend to transfer it to other open urban-data domains —mobility, air quality and seismic risk— where the same combination of dispersed official files and absent query tooling recurs.

Declaration on the Use of Generative AI

The authors used a generative AI assistant as a writing and reviewing aid, limited to language editing of the English text, formatting of the L^AT_EX source, and suggestions for restructuring the argument in response to the reviewers’ comments. The system design, the ETL implementation, the data processing, the experiments and their interpretation were carried out by the authors, who take full responsibility for the content of the article, verified every cited reference against its original source, and confirm that no text was reproduced from third-party works.

Disclosure of Interests

The authors declare that they have no conflicts of interest relevant to the content of this article.

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